

Team Wing Layout R01

Student worksheet | Tuesday September 29 2026 | Layout before coupon fabrication

Define the course baseline as a dimensioned semi-wing and trace one load to the root. This sheet records an initial design, not approval to print wing modules.

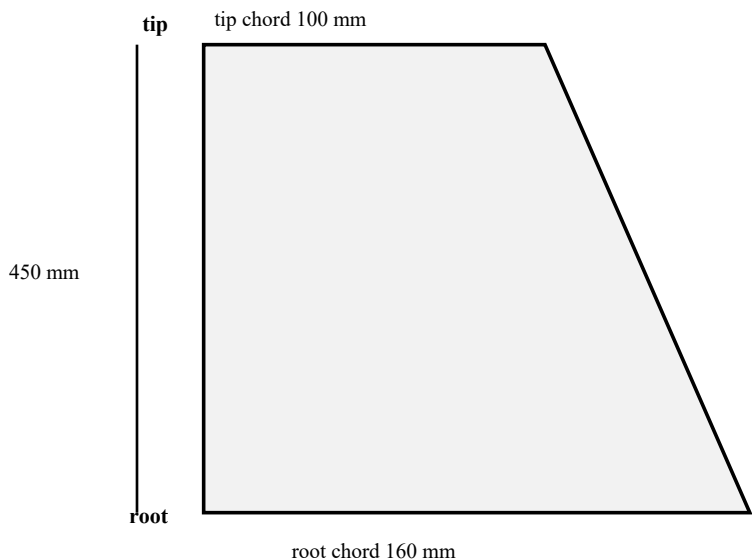
Team ID	Three team members	Instructor approved NACA code

Course baseline and design boundaries

Physical artifact	One non-flying semi-wing
Planform	s = 450 mm; root chord = 160 mm; tip chord = 100 mm
Construction	3 printed modules; skin = 1.2 mm; rib = 1.6 mm
Rod paths	Two course-issued 4 mm rods at local x/c = 0.30 and 0.60

A Top view of the project semi-wing

On the pale wing, mark the two module seams, the two rod paths, ribs, root and tip. Measure x aft from the local leading edge; label all distances in mm.



- Add by hand:**
- 1 Seam lines at y = 150 and 300 mm
 - 2 Both rod paths at x/c = 0.30 and 0.60
 - 3 Ribs and the root attachment
 - 4 One arrow for an upward surface load

The printer uses CAD from the course notebook.
This sketch is a dimension and load-path check,
not a printable model.

B Root section and interface sketch

Sketch an airfoil section in the space below. Label skin, internal rib, two rod sleeves, and the 4 mm rods. Leave the final hole clearance blank until the physical coupon has been tested.

Leading edge	Trailing edge

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Student worksheet | Engineering checks and Coupon Only notebook run

C Preliminary dimensions

Use the linear chord law $c(y) = 160 + (100 - 160)y/450$, with y and c in mm. The full area and aspect-ratio derivation will be completed on October 1.

Chord at seam $y = 150$ mm

Chord at seam $y = 300$ mm

Module length in baseline

Rod centers at root $x/c = 0.30$ and 0.60

Rod centers at tip $x/c = 0.30$ and 0.60

D Forces and the load path

A pressure load acts on the outer surface. Write a plausible path from that load to the root. At one outboard cut, identify vertical shear V_z , bending M_x , and possible torsion T_y . Do not claim that this sketch proves strength.

One structural question we must still check

E Predict before running code

For comparison only, change the tip chord from 100 to 90 mm while holding semi-span and root chord fixed. The production baseline remains 100 mm.

Predicted full-wing area direction

Predicted aspect-ratio direction

Reason for both predictions

F Coupon Only notebook record

Save a copy of the project notebook. Use the baseline values and revision R01; select Coupon Only. Do not mark the physical coupon as confirmed today.

☐ Team form saved in our Colab copy

☐ Prediction entered before analysis

☐ R01 coupon ZIP downloaded

☐ Supervised coupon slot requested

R01 coupon ZIP filename or specific blocker

Keep COUPON_CONFIRMED = False and COUPON_FIT_RESULT = Not tested yet until the rod and printed coupon have been physically tested with staff.